# Santiago Montaña

**Data Scientist | Data & AI Engineer | Conversational AI & Cloud Operations**

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## PROFESSIONAL SUMMARY

Data Scientist and Data & AI Engineer with a degree in Electronic Engineering (2024) and a Graduate Specialization in Strategic Data Analytics (UPTC, 2025). Proven track record designing and orchestrating end-to-end ETL/ELT data pipelines (Snowflake, Apache Airflow, dbt, Meltano), training predictive models and ensemble classifiers (Random Forest, XGBoost, Monte Carlo quantitative simulations tracked in MLflow), building production conversational AI & autonomous agents (LLMs, Model Context Protocol - MCP, ElevenLabs voice AI), and provisioning reproducible cloud infrastructure on AWS via Terraform (IaC), Docker, and FastAPI under CI/CD workflows.

## PROFESSIONAL EXPERIENCE

### Data Scientist & AI Developer

Jan 2026 – Present

*CIM DATA IA — Consultoría de Datos e Inteligencia Artificial — Bogotá / Remoto, Colombia*

- Trained and deployed predictive models and ensemble classifiers in Python (scikit-learn, Random Forest, XGBoost) tracked with MLflow, improving operational forecasting and prioritization accuracy by +20%.
- Engineered conversational AI solutions and autonomous agents integrating Model Context Protocol (MCP), automating enterprise workflows and reducing manual operational intervention by ~35%.
- Provisioned and standardized cloud infrastructure on AWS (EC2, S3, IAM) using Terraform (IaC), achieving 100% environment reproducibility and cutting deployment lead times by ~45%.
- Executed quantitative Monte Carlo simulations (+10,000 scenarios) and stress tests to validate data architecture resilience, orchestrating automated ETL/ELT pipelines with Python, SQL, Snowflake, dbt, and Apache Airflow under CI/CD.

### Data & AI Development Associate

May 2025 – Dec 2025

*CIM DATA IA — Soluciones Financieras y Conversacionales — Bogotá / Boyacá, Colombia*

- Analytically tuned and optimized banking conversational AI models and NLP intent pipelines, increasing intent classification accuracy by +22% and reducing fallback rates in client interactions.
- Co-developed a low-latency AI voice assistant for automated loan verification in the financial sector using Python, ElevenLabs, and third-party REST APIs, cutting verification cycle time by ~40%.
- Automated recurring operational data workflows and standardized reproducible Python environments across the engineering team, reducing environment setup errors by ~50%.

### Data & Industrial Operations Automation Intern

May 2024 – Nov 2024

*HOLCIM S.A. — Operaciones Industriales y Planta — Nobsa, Boyacá, Colombia*

- Automated industrial plant data ingestion, cleaning, and validation using Python scripts and Google Apps Script, reducing manual reporting effort by ~40%.
- Designed and deployed a centralized relational database to track +500 industrial assets and maintenance records, eliminating data silos across plant operations.
- Built interactive executive dashboards in Looker Studio to monitor real-time plant productivity, equipment availability (OEE), and operational KPIs for plant leadership.

## TECHNICAL SKILLS

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|---|---|
| **Data Engineering & Orchestration:** | Snowflake (Cloud Data Warehouse), Apache Airflow (DAG Orchestration), dbt (SQL Transformations & Data Modeling), Meltano (ELT Extraction), Pipelines ETL / ELT & Data Quality |
| **Machine Learning & Data Science:** | Scikit-learn (Random Forest, XGBoost, Ensembles), Simulaciones Cuantitativas (Monte Carlo), MLflow (Experiment Tracking & MLOps), Análisis Estadístico & Validación de Hipótesis, Feature Engineering, Pronóstico & Clasificación |
| **Conversational AI & MCP Agents:** | Model Context Protocol (MCP) Integration, Agentes Autónomos de IA & Workflows, LLMs (Claude, OpenAI, Prompt Engineering), ElevenLabs (Asistentes de Voz de Baja Latencia), Optimización de Chatbots Bancarios & NLP |
| **Cloud, DevOps & IaC:** | Terraform (Infrastructure as Code - IaC), AWS (EC2, S3, IAM), FastAPI & Pydantic (APIs REST en Producción), Docker & Entornos Reproducibles, CI/CD con GitHub Actions & Git |
| **Programming Languages:** | |

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|---|---|
| | Python (pandas, NumPy, SciPy, scikit-learn, Matplotlib), SQL (Consultas Analíticas, CTEs, Window Functions, Optimización), JavaScript / TypeScript (ES6+) |
| **Databases & Warehousing:** | Snowflake, PostgreSQL, MySQL, SQLite |
| **Visualization & BI:** | Looker Studio, Power BI, Tableau, Storytelling con Datos & KPIs Operativos |

## FEATURED PROJECTS

**Supply Chain & Marketplace Analytics (Coomproriente)** *Stack: Python, pandas, SQL, Data Visualization, Operations Analytics*

Quantitative modeling of carriers, merchants, and pricing dynamics to optimize regional supply chain logistics and marketplace operations.

**Autonomous News & Media AI Agent** *Stack: Python, LLMs, MCP, Autonomous Agents, Automation*

Autonomous AI pipeline integrating LLMs and MCP workflows for automated research, synthesis, and multi-channel content generation.

**Multiclass Vehicle Classification API & Web App** *Stack: Python, scikit-learn, FastAPI, Pydantic, Streamlit*

End-to-end classification model serialized and deployed in production with FastAPI, Pydantic schema validation, structured logging, automated tests, and Streamlit UI.

**USD/COP Exchange Rate (TRM) & Crypto Monte Carlo Predictor** *Stack: Python, Scikit-learn, Monte Carlo, NLP Sentiment, Time Series*

Predictive modeling combining historical time series, real-time news NLP sentiment analysis, and Monte Carlo stochastic simulations (+10,000 paths) to estimate market scenarios.

## EDUCATION

**Specialization in Strategic Data Analytics**
*Universidad Pedagógica y Tecnológica de Colombia (UPTC) · Graduate Specialization · Graduated 2025*

**Bachelor of Science in Electronic Engineering**
*Universidad Pedagógica y Tecnológica de Colombia (UPTC) · B.Sc. Degree · Graduated 2024*

## LANGUAGES

**Spanish** — Native (C2)

**English** — Upper-Intermediate (B2 Professional)

## CERTIFICATIONS & COURSES

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| MLOps & Despliegue de Modelos de Machine Learning | 2025 | Data Scientist con Python | Apr 2025 |
| Decision Trees y Random Forest con scikit-learn | Mar 2025 | Tableau — Visualización de Datos y Storytelling | Mar 2025 |
| Power BI para Analítica de Negocios | 2025 | Regresión Lineal con Python y scikit-learn | Feb 2025 |
| Arquitectura y Proyectos de Ciencia de Datos | Feb 2025 | Git, GitHub Profesional & Flujos CI/CD | Jan 2025 |
| Bases de Datos Relacionales y Consultas con SQL | Sep 2024 | Python Aplicado a Ciencia de Datos | Sep 2024 |